

CSAP Rubric for High-Quality Scientific Feasibility Claims

Purpose: Automated scientific feasibility systems take as input a scientific or technological capability claim and output:

- A feasibility score: [-2, -1, 0, +1, +2]
- A supporting explanation for the score
- Evidence, reasoning, or experimental results supporting the judgment

Feasibility scores are defined as:

Score	Meaning
-2	Infeasible
-1	Likely infeasible
0	Unknown / insufficient evidence
+1	Likely feasible
+2	Feasible

The purpose of the CSAP Rubric is not to judge whether a claim is feasible. Its purpose is to judge whether a claim is high quality for evaluating scientific feasibility systems.

A high-quality claim should be:

- Clear enough that different SMEs would interpret it similarly
- Specific enough to guide decomposition and evidence gathering
- Amenable to automated analysis, experimentation, simulation, or quantitative grounding
- Nontrivial, requiring projection across science or over time rather than simple lookup from existing literature

Claims should not be so vague that they cannot be assessed, nor so obvious that they can be answered directly from existing scientific consensus.

CSAP Scoring Function: Each claim is scored from 0 to 5 on four criteria: Clear, Specified, Automatable, Projective. The total score ranges from 0 to 20.

Total Score:

- 17–20 Excellent claim for feasibility-system evaluation
- 13–16 Good claim; minor improvements needed
- 9–12 Moderate claim; usable but underspecified or weakly testable
- 5–8 Poor claim; major ambiguity or automation gaps
- 0–4 Not suitable for feasibility-system evaluation

Criterion 1: Clear

Definition: The claim is written in a way that minimizes ambiguity and supports consistent interpretation by different human SMEs and automated feasibility systems. A clear claim identifies the core scientific or technological assertion without relying on vague, subjective, or overloaded language. Claims should state a single primary capability or feasibility assertion and avoid bundling multiple unrelated claims into one sentence. Unless defined, claims should avoid ambiguous terms such as “better,” “advanced,” “robust,” “safe,” “effective,” or “scalable.”

Criterion 2: Specified

Definition: The claim contains enough technical detail to support decomposition into verifiable subclaims, while still requiring meaningful reasoning and analysis. A specified claim should, as relevant, identify the target capability or scientific outcome, system, model, algorithm, or process being assessed, application context, operating domain and conditions, quantitative thresholds, benchmarks, measurable success criteria, and relevant constraints (e.g., time, scale, environment, data regime, safety limits, resource assumptions). Specificity does not mean the claim must include every experimental detail; rather, it should provide enough structure for a feasibility system to know what must be assessed. Both low-specificity and high-specificity claims can be valuable, but they test different capabilities. A lower-specificity claim may be useful if it requires the feasibility system to perform substantial decomposition. A higher-specificity claim may be useful if it enables more direct experimentation, testing, simulation, benchmarking, or quantitative analysis.

Criterion 3: Automatable

Definition: The claim is formulated so that an automated AI feasibility system could gather evidence, perform analyses, run code, conduct simulations, query databases, reproduce experiments, or otherwise generate quantitative grounding relevant to the feasibility judgment. The claim should be assessable by more than literature summarization alone. Claim can be decomposed into subclaims that map to evidence-gathering or computational tasks. Claim supports quantitative evaluation through benchmarks, datasets, simulations, calculations, models, or experiments. Claim allows an AI system to produce new evidence or intermediate results, not merely retrieve existing conclusions. A weakly automatable claim requires purely subjective expert judgment, depends on inaccessible proprietary information, lacks measurable outputs, can only be answered by broad literature review, or is framed as a policy, preference, market, or normative claim rather than a scientific feasibility claim.

Criterion 4: Projective

Definition: The claim requires a nontrivial feasibility judgment that goes beyond simply confirming what is already known. A projective claim asks whether a capability that is not yet fully demonstrated is scientifically plausible under stated or inferable conditions. Claim is not already settled by existing consensus. Claim is neither obviously impossible nor trivially already achieved. Claim requires reasoning from current evidence to a plausible

but undemonstrated capability. Claim requires assessing current technical gaps, missing enablers, or scientific advances that would change the feasibility judgment. Claim should require scientific feasibility systems to project either across scientific domains (i.e., extend known scientific principles, methods, or results into a new domain, model class, operating regime, or application context where the capability has not yet been demonstrated; a method demonstrated in domain X could enable capability Y in domain Z if capability gaps A, B, and C are resolved) or forward in time (i.e., capability is not feasible or demonstrated today but may become feasible in the future as enabling technologies mature; a capability Y could become feasible within future conditions C if advances in instrumentation, algorithms, data, manufacturing, compute, theory, or experimental methods occur).